A

The selection problem defines biological and analytical axes

INTRINSIC SUBSTRATE

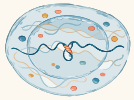
Can this transcript form a suitable duplex?



sequence · predicted fold
genomic organization · burden

CONTEXT / REALIZATION

Does that substrate exist in the relevant state?



tissue variation · RBP / abundance
transcript / repeat · chromatin

TARGET-LOCAL SELECTION

Which adenosine is selected within the duplex?



target geometry · focal state
aligned local neighborhood

STATISTICAL DEPENDENCE Sites within one duplex are dependent; label recovery and pair-first weighting were separated.

AI / MODELING SEARCH more supervision · cross-context sharing / multitask learning · richer representations · capacity / integration

B

A common test turns candidate information into an evidence map

BIOLOGICAL SCALE	SEQUENCE–STRUCTURE REFERENCE	PAIR-DISJOINT DEVELOPMENT OOF	SAME-FOLD + MATCHED CONTROL	EFFECT + UNCERTAINTY STATUS + STOP RULE
	CANDIDATE INFORMATION AND RESULT			MAP UPDATE
INTRINSIC SUBSTRATE	● Sequence + predicted fold	● Geometry → pair burden	× Broad repeat / transcript panel	**whole-duplex signal + between-substrate propensity**
CONTEXT / REALIZATION	● Same-site tissue variation	× RBP × bulk abundance	▲ Bulk H3K36me3 ○ Matched donor / cell / in vivo state	**context dependence visible; responsible state unresolved**
TARGET-LOCAL SELECTION	▲ Contact geometry + focal node	● Aligned local representation	× Shifted matched-capacity control	**target-centered local signal recovered**
STATISTICAL DEPENDENCE	● Additional valid high / low labels	● Pair-first weighting		**label recovery and weighting make separable gains**
AI / MODEL SUFFICIENCY	▲ Graded objectives + shared tissues	× Foundation representations	× Capacity + joint decoding ● Fixed prediction integration	**targeted changes help; generic complexity yields little more**

● supported / selected ▲ limited × not supported in tested formulation ○ matched data unavailable

C

The completed search changes the next question

RECOVERABLE

Whole-duplex sequence–structure
Pair geometry at substrate level
Aligned target-local information
Pair-aware statistical weighting

PARTIALLY RESOLVED

Tissue-associated variation is visible
Bulk context adds no stable residual site signal
Generic model expansion shows diminishing returns

MEASURE NEXT

Editing paired to donor and cell state
Transcript processing and regulator occupancy
ADAR state and in vivo RNA conformation

FROZEN SYNTHESIS CHECK The selected principles improved AUROC, AUPRC, F1, and Brier score in all five held-out endpoints.
unseen benchmark pairs · external-cohort transport remains untested